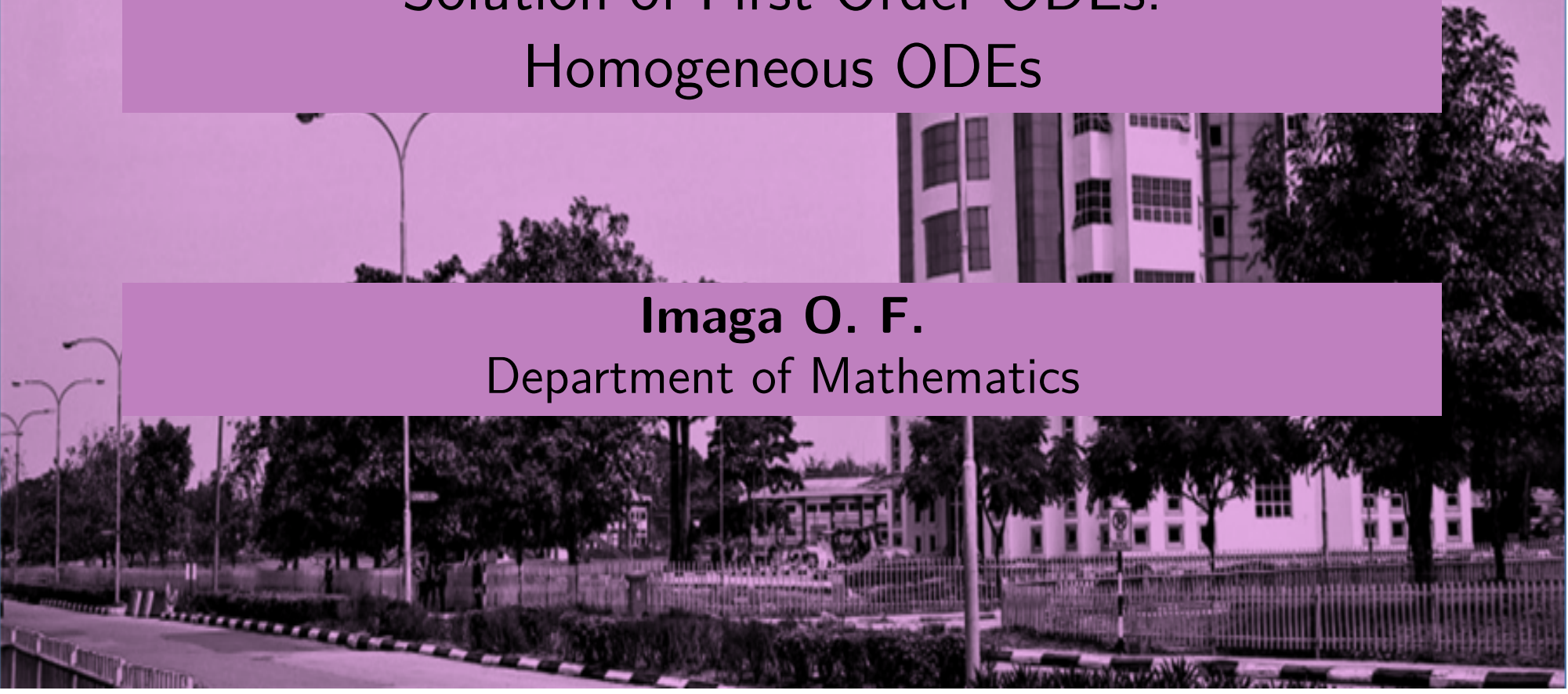
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# MTH213/MAT213: Differential Equation I

## Solution of First Order ODEs: Homogeneous ODEs



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# Introduction

A first order ODE of the form

$$\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)} \qquad 1$$

is said to be homogeneous if the degree of each term of $f(x,y)$ and $g(x,y)$ is the same.

. Examples of homogeneous ordinary differential equation 1.

$$\frac{dy}{dx} = \frac{xy + y^2}{x^2}$$

2. $2xy\frac{dy}{dx} = x^2 + y^2$

# Introduction Contd.

A differential equation of the form

$$\frac{dy}{dx} = F(x, y) \qquad 2$$

is said to be homogeneous if $F(x, y)$ is a homogeneous function of degree zero i.e.

$$F(\lambda x, \lambda y) = \lambda^0 F(x, y) = F(x, y).$$

**Example:** Show that the ordinary differential equation $\frac{dy}{dx} = \frac{y}{x} + x\sin\frac{y}{x}$ is a homogeneous ordinary differential equation.

**Solution:**

$F(x, y) = \frac{y}{x} + x\sin\frac{y}{x}$ then $F(\lambda x, \lambda y) = \frac{\lambda y}{\lambda x} + \lambda x\sin\frac{\lambda y}{\lambda x} \neq F(x, y).$

# Introduction Contd.

Let's look at another test for homogeneity, where we use the substitution $y = vx$. Given the differential equation:

$$\frac{dy}{dx} = \frac{y}{x} + x\sin\left(\frac{y}{x}\right).$$

Let $v = \frac{y}{x}$, so $y = vx$.

Then, by the product rule: $\frac{dy}{dx} = v + x\frac{dv}{dx}$

Substitute $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$ into the original equation:$v + x\frac{dv}{dx} = \frac{vx}{x} + x\sin\left(\frac{vx}{x}\right) v + x\frac{dv}{dx} = v + x\sin(v)$.

Now, subtract $v$ from both sides: $x\frac{dv}{dx} = x\sin(v)$

Divide by $x$: $\frac{dv}{dx} = \sin(v)$

# Introduction Contd.

The resulting differential equation is a separable equation. Since the original differential equation can be transformed into a separable equation using the substitution $y = vx$, it confirms that it is a homogeneous ordinary differential equation.

Exercise: Show that the ordinary differential equation $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x$ is homogeneous.

# Steps to Solve a Homogeneous ODE

Use the following steps to solve a Homogeneous ODE:

**Step 1:** Set $y = vx$ so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$. Substitute these into the given ODE.

**Step 2:** Separate the variables:

$$f(y)dy = g(x)dx \tag{3}$$

**Step 3:** Integrating both sides of equation (3):

$$\int f(y)dy = \int g(x)dx, \tag{4}$$

**Step 3:** Add an arbitrary constant C on the right hand side.

**Step 4:** Replace $v$ by $\frac{y}{x}$ and simplify.

# Examples

1. Show that the ordinary differential equation $(2xy + x^2)y = 3y^2 + 2xy$ is homogeneous and solve it.

**Solution**

The equation can be rewritten as $\frac{dy}{dx} = \frac{3y^2 + 2xy}{2xy + x^2}$

The ODE is homogeneous because each term in the equation is of degree two.

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Set $y = vx$ so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$. Substitute these into the given ODE

$$v + x\frac{dv}{dx} = \frac{3v^2x^2 + 2vx^2}{2v^2x^2 + x^2} = \frac{3v^2 + 2v}{2v + 1}$$

# Examples Contd.

$$\Rightarrow x\frac{dv}{dx} = \frac{3v^2 + 2v - 2v^2 - v}{2v+1} = \frac{v^2 + v}{2v+1}$$

Separating variables yields: $\left(\frac{2v+1}{v^2+v}\right) dv = \frac{dx}{x}$.

Integrating both sides: $\int \left(\frac{2v+1}{v^2+v}\right) dv = \int \frac{dx}{x}$

$$\Rightarrow \log(v^2 + v) = \log x + \log c \Rightarrow v^2 + v = cx$$

Replace $v$ by $\frac{y}{x}$: $\frac{y^2}{x^2} + \frac{y}{x} = cx \Rightarrow y^2 + xy = cx^3$

# Exercises

1. Solve the following ODEs

a. $(y^2 - xy)dx + x^2 dy = 0$

b. $2xy^2 dy - (x^3 + 2y^3)dx = 0$

c. $(x^3 - 3xy^2)dx + (y^3 - 3x^2 y)dy = 0, \quad y(0) = 1$

d. $\left(x\cos\frac{y}{x} + y\sin\frac{y}{x}\right)y - \left(y\sin\frac{y}{x} - x\cos\frac{y}{x}\right)x\frac{dy}{dx} = 0$

2. Show that the differential equation $\left(1 + e^{\frac{x}{y}}\right)dx + \left(y^2 - x^2\log\frac{x}{y}\right)dy = 0$ is a homogeneous ordinary differential equation and find the particular if $y(1) = 0$.